

Photoacoustic Sensing in Sustainable Mining: Predicting Coal Susceptibility to Spontaneous Combustion

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Abstract— Spontaneous combustion of coal is a major concern in the coal mining industry leading to economic loss, environmental hazard and workers' life risk. The coal in the stockyard is highly vulnerable to weather conditions; hence, it must be monitored to initiate preventive measures before the coal ignition begins. This paper presents a novel way to classify coal based on its spontaneous combustion susceptibility through a compact photoacoustic sensor. Here, the coal pellets are prepared, and the time domain photoacoustic signal is acquired indicating intrinsic thermal properties of a coal. The derived time domain feature is used to train and test the machine learning (AI) model. In total, 360 data points from 45 coal pellets are used for the experiments. The model's accuracy to classify the coal into three classes, e.g., high, moderate and less susceptible to spontaneous combustion is obtained as 81.9%. Before testing, the model is validated for its overfitting and data distribution using Kernel density estimation.

Keywords— Photoacoustic sensing, Spontaneous combustion of coal, thermal properties.

I. INTRODUCTION

Coal plays a pivotal role in global electricity generation and further is an inherently indispensable component for several industries including iron and steel. Thus, safe mining and storage of coal is of utmost importance to the mining industries [1]. The coal's susceptibility towards spontaneous combustion (SC) possesses a serious challenge in its storage and transportation. The spontaneous combustion of coal does not only produce toxic greenhouse gases but also leads to fire fatalities and huge economic losses [2].

Conventionally, the SC susceptibility of coal is determined through laboratory based thermal testing methods such as crossing point temperature (CPT), thermogravimetric analysis (TGA), differential scanning calorimetry (DSC), etc. [3]. In the recent past, several artificial intelligence (AI) tools were used with the dataset obtained from conventional methods to classify coals as highly susceptible, moderately susceptible or less susceptible towards SC [4]. This classification of coal is extremely important as it helps the mine management to prioritize the transportation and usage of different coal stocks. Some other attempts to predict the susceptibility of coal towards SC such as gas (CO , CH_4) sensing, firefighting chemicals, water treatment are made

but could not succeed [5]. While gas sensing is possible only when the ignition is initiated, the conventional methods suffer tedious and time-consuming measurements. Moreover, the coal's intrinsic properties change significantly with external factors thus, the continuous monitoring of stockpiles is required for predicting SC susceptibility [6]. By regular on-site monitoring the coal's properties and detecting any signs of spontaneous heating early on, the mine management can implement timely interventions to prevent further heating and mitigate the risk of spontaneous combustion. Thus, there exist a strong need of a rapid, in-situ sensing method for continuous monitoring of coal stockpiles to predict SC susceptibility of coal in its early onset (incubation period).

This study proposes a novel way to determine the thermal properties of coal through a compact, rapid and field deployable photoacoustic (PA) sensing and thereby predicting its susceptibility towards SC in its incubation period. Briefly, photoacoustic sensing is a pump-probe technique wherein nano-second laser pulses are used to irradiate the sample. The sample upon absorption of the laser pulse undergoes temperature excursion and thereby thermoelastic expansion. Subsequently, the sample releases the gained energy in a non-radiative manner which is the pressure wave or a photoacoustic wave [7-9].

In this study, due to the low signal to noise ratio (SNR) of the time-domain PA signals from different coal samples the Empirical Mode Decomposition (EMD) algorithm is employed, and the different modes are analyzed to differentiate the coal samples. Different coal samples are collected and TGA analysis is performed to obtain the ground truth of the thermal properties. Further, these samples are classified into three different classes namely high susceptibility, moderate susceptibility and low susceptibility towards SC. The three classes are defined by using K-means clustering of the ground truth values. For each class of a coal sample, 120 data points of PA signal are acquired out of which 80% are used to train the machine learning (ML) model with the known label and the remaining 20% for testing the model. The model is also validated by observing the probability distribution of the

data points into three classes using Kernel density estimation.

II. MATERIAL AND METHODS

A. Sample preparation

The coal samples are collected from the stockyards which are extensively distributed across the India namely Nagpur, Dhanbad and Talcher area. The sites are strategically selected to ensure variation in spontaneous combustion susceptibilities of samples necessary to prepare a balanced dataset. To acquire the PA signal from the coal, it is used in the pellet form. For making the coal pellets, the raw coal is crushed, grinded and sieved to get a coal powder of a specific particle size i.e. 212 μ m. Subsequently, the powder is mixed with the resin and hardener in appropriate ratio (2:1) and stirred well thereafter. As a standard practice for coal pellet preparation, the total mass of resin and hardener is equal to the mass of the coal sample. For example, for 5g of coal powder, 2.5ml of resin and 1.25ml of hardener is used. Further, this mixture is poured into the mold and allowed to settle for approximately 90 min to get the pellets. The thickness and diameter for the pellets are kept as 6mm and 40mm respectively.

B. Experimental setup

For experiments, pulsed laser diode module (905D4S16S) from Laser components (wavelength 905nm) is used for excitation. The focused laser beam with a diameter of 2mm and a pulse width of 400ns is used to irradiate the pellet. The PA wave generated in the pellet is acquired using the ultrasound sensor, V-303 SU (coupled using acoustic gel) from Olympus Inc. The PA signal from the ultrasound sensor is amplified using the custom-made pre-amplifier having a constant gain of 40dB and bandwidth of 5 MHz. Due to the low SNR, the PA signal averages 100 times, and the final averaged signal is stored in the computer through the data acquisition system (DAQ). Laser excitation and data acquisition are synchronized through the trigger pulse. Fig.1 shows the experimental setup.

C. EMD analysis and feature selection

EMD analysis is performed with the PA signal due to its non-stationary and non-linear characteristics. Using EMD, a PA signal is decomposed into its constituent set of intrinsic mode function (IMF) which are approximately 8 IMFs for each PA signal. Each IMF represents an oscillatory response with the varying frequency over time. Since the initial IMFs (IMF-1,2, etc.) contain the highest frequency components, the noise involved with these components is significant which makes the analysis difficult. Therefore, in this study, IMF3 is used for analysis. The time (T_p) corresponding to the peak amplitude in the IMF3 of a PA signal is used as a feature for coal classification. The shift in T_p for different coal samples can be attributed to the change in its acoustic wave velocity and the rise time (mechanical properties of the sample).

D. TGA of different coal samples and coal classification

The thermal properties of different coal samples are measured using the TGA.

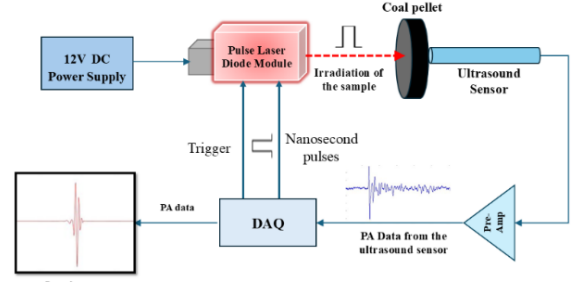


Fig.1: Experimental setup

Fig.2 shows the TGA and derivative of TGA (DTG) plot for three different coal samples and highlights their ignition temperature (T_{ign}). If W is the TGA function, then DTG is obtained by (dW/dt) and T_{ign} is given as temperature corresponding to the 10% fall of the peak of dW/dt . The detailed explanation on TGA can be studied from the referred paper [10].

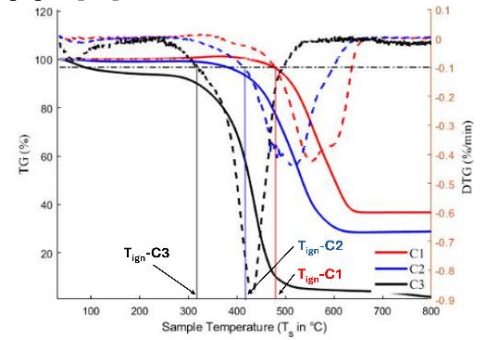


Fig.2: TGA/DTGA curve for three different coal samples

Literature reveals that the ignition temperature (T_{ign}) obtained from DTGA plot could be the best index for coal classification towards SC susceptibility [11]. Thus, T_{ign} of different coal samples are used with the K-means clustering algorithm to divide the total (obtained from seven different locations) coal samples into three classes (Less, Moderate and High susceptibility) as shown in Table 1. For each class of coal, a total of 120 data points is used in analysis.

TABLE 1: K-MEANS CLUSTERING OF COAL SAMPLES

Sample Code	Ignition Temperature (T_{ign}) in °C	Cluster centroid (°C)/ Susceptibility class
C1, C2	465.05, 476.51	470/ Less
C3, C4	412.297, 439.72	426/ Moderate
C5, C6, C7	317.61, 346.88, 346.77	337/ High

III. RESULTS AND DISCUSSION

The time domain PA signals for three different coal samples (C1-C3) are shown in Fig.3a. As discussed, T_p needs to be measured from the PA signal in Fig.3a to determine the thermal properties of a coal. However, the low SNR of PA signal poses a challenge to identify the single peak to obtain T_p . Thus, the EMD algorithm is used with the PA signal and

Fig.3b shows the IMF3 response for the PA signals corresponding to Fig.3a. With the removal of high frequency noise, the IMF3 response shows a distinct peak of the PA signal from where T_p can be identified. The three samples (C1-C3) shown in Fig.3b, respectively belongs to the three different classes (less, moderate and high) of coal's susceptibility to SC.

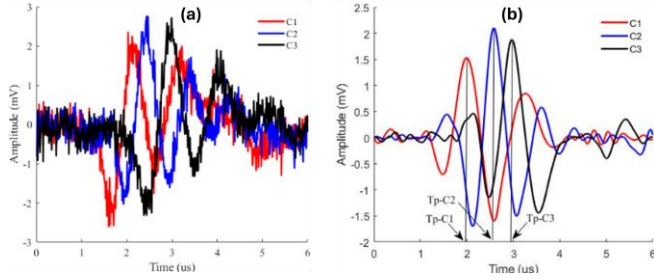


Fig.3: For three different coal samples- (a). PA signals, (b). IMF3 response

Fig.3b reflects that ($T_p-C1 < T_p-C2 < T_p-C3$) which means the velocity (inversely related to T_p) of PA wave for less susceptible class is higher while for the higher susceptible class the velocity is less. Moreover, literature reveals that the acoustic velocity is directly proportional to the specific heat capacity and thermal conductivity of the sample [12,13]. Also, the higher thermal conductivity suggests that the heat dissipation from the coal sample is more and hence, it exhibit lesser susceptibility towards SC [14]. Hence, acoustic velocity or T_p can be used to depict the thermal properties and thereby susceptibility of the coal towards SC. Further, 360 data points from the three classes of coal samples are acquired and their T_p values are obtained. Fig.4a shows the empirical probability distribution of the T_p values corresponding to the three classes of coal samples. The probability distribution is obtained using the Kernel density estimation (KDE) and it shows that the distribution of T_p values for three classes is distinct with the minor overlap.

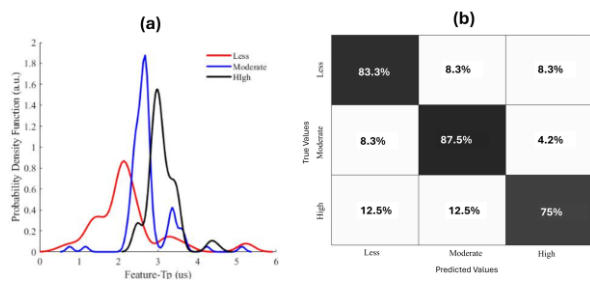


Fig.4: a. Empirical probability distribution of feature (T_p values), b. Confusion matrix

To predict the class of a coal from the PA response, an ML model specifically is trained and tested. Table 2 shows the comparison of different classification models to predict spontaneous combustion susceptibility of coal. It is reflected from Table 2 that Ensemble-Boosted trees classifier has outperformed in comparison to the other classifiers. Therefore, this study utilize the Ensemble Boosted trees classifier for predicting the coal susceptibility towards SC.

The training accuracy of the model is confirmed by the K-fold validation method. For 3, 5 and 10 folds, the accuracy is found to be 85.8%, 86.6% and 83.8% respectively. Fig. 4b shows the confusion matrix for the tested samples.

TABLE 2: CLASSIFICATION MODEL COMPARISON

Classification Model	Accuracy	Precision	Recall	F1-score
Wide Neural Network	79.2%	79.2%	79.2%	79.2%
Fine KNN	80.6%	82%	80.6%	81.29%
Ensemble Boosted Tree	81.9%	80.18%	81.9%	80.42%

From Fig.4b, the average prediction accuracy to classify the coal samples into three classes is found to be 81.9%. It is evident from Fig.4b that for class 3 the accuracy is low compared to the other two classes. The reason could be the heterogeneity of coal samples of class 3 which would have exhibited the characteristics of class 1 and class 2 samples. The false prediction of class 3 can be visualized from Fig. 4a where the overlap of class 1 and class 2 into class 3 is observed. The classification accuracy can further be improved by training the model with a more robust dataset involving the coal samples with varying thermal properties. This method can be applied to monitor the coal stockpile by further reducing the pellet preparation time or directly using the raw coal samples in the experimental setup. Table 3 shown below demonstrates the efficacy of the proposed PA sensing method when compared with the existing techniques to predict SC susceptibility of coal.

TABLE 3: COMPARISON OF DIFFERENT TECHNIQUES

Techniques	Period of SC	Fast response time	Compact system	Cost effective
Thermal Imaging	Indication	Yes	Yes	No
Multi-gas Sensing	Indication	Yes	Yes	Yes
CPT, TGA/DSC	Incubation	No	No	No
FTIR Spectroscopy	Incubation	Yes	No	No
Photoacoustic Sensing	Incubation	Yes	Yes	Yes

IV. CONCLUSION

The photoacoustic sensing technique with AI is implemented to classify the coal sample towards their spontaneous combustion susceptibility. To our best, this is the first time any such compact and field deployable sensing methods is used to predict the SC susceptibility of coal. Going forward a compact photoacoustic sensing system with an edge machine can be envisaged for the in-situ prediction of coal's susceptibility towards spontaneous combustion. This would lead to minimizing coal loss and preserving inventory quality whilst minimizing the toxic greenhouse gases emissions, hence, aiding sustainable mining.

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